

Chapter 38: Contraception — Revision Notes

Williams Obstetrics 26e, pp. 1714–1758 of the PDF. High-yield numbers are in **bold**. The seven tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 3 figures are included from the PDF (Figure 38-1 appears twice in the PDF, on pp. 1724 and 1725; it is included once).

Big picture: Nearly **half of US pregnancies are unintended**. **LARC (IUDs and the implant) is top-tier** (<1% failure, perfect = typical use) and should be encouraged for suitable women. The **US MEC** grades each method **1–4** for each condition. After delivery, **estrogen is avoided for the first 21 days**, while progestin-only methods and IUDs can start immediately.

1. Postpartum Context, Efficacy and Eligibility

The puerperium

- **Exclusive breastfeeding** → **ovulation unlikely in the first 10 wk**. Not reliable if the infant nurses only by day.
- **Ovulation usually precedes the first menses** → waiting for menses risks pregnancy. **After the first menses, contraception is essential**.

Table 38-1. Contraceptive Failure Rates of Reversible Methods During the First Year and Use Rates^a

Method	Perfect Use	Typical Use	Percent Use ^b
Top tier (most effective)			
Intrauterine devices:			11.8
52-mg LNG-IUS	0.1	0.1	–
T380A Cu-IUD	0.6	0.8	–
Etonogestrel implant	0.1	0.1	2.6
Female sterilization	0.5	0.5	21.8
Male sterilization	0.1	0.15	6.5
Second tier (very effective)			
Combination pill	0.3	7	24.9
Vaginal ring	0.3	7	2.4
Patch	0.3	7	0.2
DMPA	0.2	4	3.9
Progestin-only pill	0.3	7	0.4
Third tier (effective)			
Condom			14.6
Male	2	13	–
Female	5	21	–
Diaphragm + spermicides ^c	16	24	–
Fourth tier (least effective)			
Spermicides ^c	18	28	–
Sponge ^c — multiparas	20	27	–
Sponge ^c — nulliparas	9	14	–

Values are percentages of women with a pregnancy in the first year. ^aAmong women in the United States using contraception. ^bThis sum totals less than 100% as withdrawal (8.1%) and natural family planning (2.2%) values are not presented in the table. ^cCombined into "other method" category, which has a use rate of 0.6%. Cu-IUD = copper-containing intrauterine device; DMPA = depot medroxyprogesterone acetate; LNG-IUS = levonorgestrel-releasing intrauterine system.

Memory hook: Typical-use failures by tier: **LARC <1 → DMPA 4 → pill/patch/ring/POP 7 → male condom 13 → spermicide 28.**

US Medical Eligibility Criteria (US MEC, CDC 2016)

- Six method groups: **LNG-IUS, Cu-IUD, implants, DMPA, POPs, CHCs** (COCs, ring, patch).

Category	Meaning
1	No restriction
2	Advantages generally outweigh risks
3	Risks generally outweigh advantages
4	Unacceptable health risk

- Contraception usually carries **less risk than pregnancy**. Sterilization is covered in Ch 39.

2. Intrauterine Devices

- **Five US IUDs**, all chemically active (copper or progestin); flexible **T-shaped polyethylene frame with barium** (radiopaque).
- LNG devices differ in size, string color, lifespan and **silver ring** (at stem–arm junction). **Smaller devices may fit a nulliparous uterus better.**
- **ParaGard (T380A):** copper wire around the stem + a copper sleeve on each arm.

Table 38-2. Properties of Intrauterine Devices

Active Agent	Quantity of Active Agent	Width × Height (mm)	Insertor Tube Diameter (mm)	FDA-approved Duration of Use (yr)	String Color	Silver Ring	Brand Name
LNG	52 mg	32 × 32	4.4	5	Tan	No	Mirena
LNG	52 mg	32 × 32	4.8	6	Blue	No	Liletta
LNG	19.5 mg	28 × 30	3.8	5	Blue	Yes	Kyleena
LNG	13.5 mg	28 × 30	3.8	3	Tan	Yes	Skyla, Jaydess
Copper	380 mm ³	32 × 36	4.4	10	White	No	ParaGard

FDA = U.S. Food and Drug Administration; LNG = levonorgestrel.

Contraceptive action

- **Prevention of fertilization is favored.**
- **Intense sterile endometrial inflammation** (especially **Cu-IUD**) → inflammatory fluid in uterus and tubes ↓ sperm and egg viability; also acts against any blastocyst.
- **LNG-IUS:** endometrial **atrophy** + **scant, viscous cervical mucus**.
- Ovulation inhibition is **inconsistent with LNG-IUS and absent with Cu-IUD**.

Method-specific adverse effects

Ectopic pregnancy

- IUDs **halve the absolute number of ectopics** vs no contraception.
- But they prevent intrauterine implantation better → **if an IUD fails, a higher proportion of pregnancies are ectopic.**

Lost device (missing strings)

- **Expulsion is most common in the first month** → check strings **4–6 wk** after insertion (after menses); the woman then checks monthly after menses.
- **Cumulative 3-yr expulsion ~10%** after nonpuerperal insertion.
- Strings not seen → expelled, perforated, malpositioned, or normal with strings curled up.
- **Stepwise approach:**
 - (1) **Exclude pregnancy**
 - (2) Twirl a **cytology brush** in the endocervical canal
 - (3) Gently probe the cavity with a **Randall stone clamp** or hooked rod
 - (4) **Transvaginal sonography** (3-D improves views)
 - (5) If inconclusive or no device seen → **plain abdominopelvic radiograph**; CT or MR as alternatives
- **⚠ Never assume expulsion unless the device was seen.** MR at 1.5 or 3 T is safe with an IUD.

Perforation

- **~1 per 1000 insertions**; risks: **puerperal insertion and breastfeeding.**
- Recognized when the sound/insertor goes **farther than expected.**
- **Acute:** usually **fundal**, little bleeding (myometrium contracts) → **observe** if no brisk bleeding from the os. Rare lateral perforation → uterine artery laceration → laparoscopy/laparotomy.
- **Occult/embedded:** pain, bleeding or missing strings → imaging.
 - Arm partly embedded → steady transcervical traction.
 - Mainly intrauterine → **hysteroscopic** removal.
 - Nearly or fully through the wall → **laparoscopic** removal (often on cul-de-sac or omentum; bowel/bladder perforation possible).

Menstrual changes

- Dysmenorrhea and irregular bleeding → **NSAIDs or tranexamic acid.**
- **Heavy bleeding mainly with Cu-IUD** → iron-deficiency anemia (oral iron).
- **LNG-IUS:** spotting for up to **6 months** → progressive **amenorrhea (~10% at 1 yr, 35% at 3 yr)**, often with less dysmenorrhea.

Infection

- **Risk greatest in the first 3 wk after insertion:** *N. gonorrhoeae*, *C. trachomatis*, vaginal flora.
- Screen women at STD risk **before or at insertion**; **don't delay insertion** for STD or Pap results in asymptomatic women. If positive later → treat, **IUD may stay.**
- **No routine antibiotic prophylaxis; no endocarditis prophylaxis.**
- After month 1, **no ↑ infection** in low-risk users; little if any ↑ infertility. **Adolescents and nulliparas at low STD risk are good candidates** (ACOG). Safe in **HIV/immunosuppression**; doesn't ↑ HIV acquisition.

Infection	Management
Mild–moderate PID	Antibiotics; IUD may be retained . Remove if no improvement in 48–72 h
Tuboovarian abscess	IV broad-spectrum antibiotics + IUD removal
Septic abortion	Immediate uterine evacuation + antibiotics
<i>Actinomyces</i> on Pap, asymptomatic	Keep IUD, no antibiotics
<i>Actinomyces</i> with infection (fever, weight loss, pain, abnormal bleeding/discharge)	Remove IUD + gram-positive coverage

Pregnancy with an IUD

- **Exclude ectopic and pelvic infection.**
- Infection → broad-spectrum antibiotics + **prompt evacuation**.
- **Intrauterine, no infection, strings visible** → **remove by gentle traction** → ↓ abortion, chorioamnionitis, preterm birth.

Cu-IUD in pregnancy (Tatum)	Abortion	Preterm delivery
Left in situ	54%	17%
Removed promptly	25%	4%

- Few LNG-IUS data (practice extrapolated from copper).
- **Strings not visible:** searching for the device may cause abortion; Williams does not use sonographic/hysteroscopic retrieval. Find and remove the IUD at delivery.

IUD insertion — timing

Table 38-3. Contraindications to IUD Use

Category	Contraindication
Both IUD types	Pregnancy or suspicion of pregnancy
	Distorted uterine cavity
	Acute PID
	Postpartum/postabortal endometritis in past 3 months
	Uterine bleeding of unknown etiology
	Acute untreated LGT infection
	Conditions linked to pelvic infection risk ^a
	Allergy to device components
Specific to Cu-IUD	Coexisting retained IUD
	Wilson disease
Specific to LNG-IUS	Known or suspected uterine or cervical cancer
	Prior PID unless a subsequent IUP has occurred
	Known or suspected uterine or cervical neoplasia
	Acute liver disease or benign/malignant liver tumor
	Known or suspected current or prior breast cancer or other progestin-sensitive cancer

^aThese include multiple sexual partners, severe immune compromise, intravenous drug use, and recent PID or endometritis. Cu-IUD = copper-containing intrauterine device; IUD = intrauterine device; IUP = intrauterine pregnancy; LGT = lower genital tract; LNG-IUS = levonorgestrel-releasing intrauterine system; PID = pelvic inflammatory disease.

Placement	Definition	Expulsion
Interval	After complete uterine involution	Lowest
Immediate	Within 10 min of placental delivery (also after miscarriage or surgical abortion, if no infection)	Slightly higher
Early	>10 min but within the first month postpartum	Highest

- Also possible **1 wk after mifepristone** once medical abortion is complete.
- **Immediate placement** → **more women end up with a retained IUD**, because many never return for interval insertion.

Table 38-4. U.S. Medical Eligibility Criteria Category of Contraceptive Methods Related to Breastfeeding and Time After Delivery

CHCs ^a	Cat.	DMPA, POPs, Implants	Cat.	LNG-IUS	Cat.	Cu-IUD	Cat.
Breastfeeding		Breastfeeding		± Breastfeeding^c		± Breastfeeding^c	
<21 d	4	<1 month	2	<10 mins	2	<10 min	1
21 d to <30 d	3	≥1 month	1	10 mins to ≤4 wks	2	10 min to ≤4 wks	2
30–42 d ^b	2			≥4 wks	1	≥4 wks	1
>42 d	2			Puerperal sepsis	4	Puerperal sepsis	4
Nonbreastfeeding		Nonbreastfeeding	1				
<21 d	4						
21–42 d ^b	2						
>42 d	1						

^aCombined hormonal contraceptive (CHC) group includes pills, vaginal ring, and patch. ^bAssociated risks that increase puerperal category score include: age ≥35, transfusion at delivery, body mass index ≥30, postpartum hemorrhage, cesarean delivery, smoking, preeclampsia. ^cWith or without breastfeeding. Cat. = category; Cu-IUD = copper-containing intrauterine device; DMPA = depot medroxyprogesterone acetate; LNG-IUS = levonorgestrel-releasing intrauterine system; POPs = progestin-only pills.

Memory hook: "21 days, no estrogen" (VTE risk; category 4 whether or not breastfeeding). **Puerperal sepsis = category 4 for any IUD.**

Postpregnancy insertion techniques

- **After 1st-trimester evacuation:** standard manufacturer technique. Larger cavity → **ring forceps under sonographic guidance**.
- **Immediately after vaginal or cesarean delivery:** by hand, inserter tube, or ring forceps; arms need **not** be folded into the tube.
 - **At cesarean:** through the unsutured hysterotomy to the **fundus**; a hand on the outer fundus gives counterpressure. Strings are directed, **not pulled**, toward the cervix.
 - **After vaginal birth (instrumented):** re-prepare the vulva and **change gloves after placental delivery, before perineal repair**. Hold the anterior cervical lip with one ring forceps; a second grasps the stem and places it at the fundus. **Keep jaws open on withdrawal** so strings aren't caught. Manual insertion also possible. Abdominal-hand counterpressure guides placement.

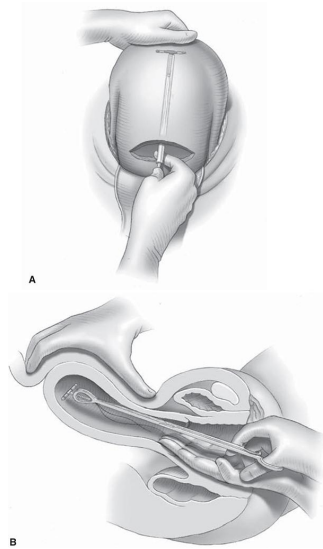


Figure 38-1. Postpartum IUD placement: (A) inserter through the hysterotomy to the fundus at cesarean, with fundal counterpressure; (B) ring forceps to the fundus after vaginal delivery, guided by an abdominal hand.

Interval insertion technique

- Easiest **near the end of menses** (cervix softer, early pregnancy excluded), but can be done **any time** if pregnancy is reasonably excluded.
- Analgesia: **lidocaine–prilocaine cream** or **paracervical block**.
- Bimanual exam (size, position, abnormalities). **Treat mucopurulent cervicitis or significant vaginitis first.**
- Antiseptic prep, sterile instruments → **tenaculum** + gentle traction straightens the canal → **uterine sound** for direction and depth → insert per package insert.
- **Trim strings to 3–4 cm** and record length. Oral NSAID for cramping.
- Suspected malposition → confirm (sonography). Not fully intrauterine → **remove and replace with a new device**. ⚠ **Never reinsert an expelled or partly expelled device.**

3. Progestin-Only Contraceptives

Actions and side effects

- **Mechanisms: LH suppression** → **no ovulation, thick cervical mucus, endometrial atrophy.**
- **Rapid return of fertility** — except **DMPA**.
- Theoretical concern that early postpartum progestin could hinder lactogenesis (progesterone withdrawal triggers it) → reflected in US MEC 2 at <1 month, although studies support safety.
- **Irregular bleeding = the commonest reason for stopping.** Options: counseling; **2-wk estrogen course** or short **NSAID** course; a few cycles of COCs with DMPA or implant. Long use → atrophy → **amenorrhea** (often welcome; less iron deficiency).
- **Generally no effect** on lipids, glucose, hemostasis, liver or thyroid function, BP. **Exception: DMPA ↑ LDL and ↓ HDL** (less desirable with CV risk).
- **No ↑ genital, liver or breast neoplasia.**
- Weight gain and fractures are not prominent — **except DMPA**.
- **Functional ovarian cysts** ↑ (usually no intervention). **No ↑ depression.**

Progestin contraindications

- **Absolute:** current **pregnancy, breast cancer, acute liver disease, liver tumors, undiagnosed abnormal genital bleeding.**

- Nexplanon and DMPA labels add current/past **thromboembolism**, but **US MEC rates progestin methods category 2** for these women.

Progestin implants

- Subdermal, inner upper arm, **~8 cm from the elbow; top-tier**.
- **Nexplanon**: single-rod **etonogestrel**, releases $\geq 30 \mu\text{g/day}$, **3 years**; **radiopaque**. Replaced Implanon (**not radiopaque**).
- **Levonorgestrel** two-rod implants (outside US; total **150 mg LNG**): **Jadelle 5 yr** (FDA-approved, not marketed in US); **Sino-implant II 3 yr**.

Implant-specific adverse effects

- Complications in **~1% of insertions and ~5% of removals**.
- Too-deep insertion or removal exploration → injury to **medial antebrachial cutaneous nerve** branches and the **median nerve**. Deep/intramuscular or near neurovascular structures → **surgeon familiar with arm anatomy**.
- **Nonpalpable implant** → **image before removal**:
 - **Nexplanon**: **2-D x-ray**.
 - **Implanon**: **10–15-MHz linear sonography** or **MR**.
 - Rare migration (e.g., lung). If still not found → **etonogestrel blood level** (via manufacturer).

Implant insertion

- **Etonogestrel implant**: **ideally within 5 days of menses onset**. LNG implants: contraceptive within **24 h** if placed in the **first 7 days** of the cycle.
- **Switching**: on the day of the **first placebo COC pill**; the day the next **DMPA** injection is due; or **within 24 h of the last POP**.
- Other times (not pregnant) → **back-up for 7 days**. Can be placed **before discharge** postpartum or postabortion.
- **Technique (Nexplanon)**:
 - Nondominant arm **abducted, elbow flexed**.
 - Mark **8–10 cm proximal to the medial humeral epicondyle** and **3–5 cm posterior to the biceps–triceps groove**; second mark **4 cm proximal** (guide along the long axis).
 - Sterile prep; **1% lidocaine** track along the path.
 - Needle bevel enters at **30°** → flatten horizontally once the bevel is in → **tent the skin** while advancing subdermally → pull the lever back to retract the needle and leave the implant.
 - **Both patient and provider palpate both ends of the 4-cm rod**. Pressure bandage, removed next day.

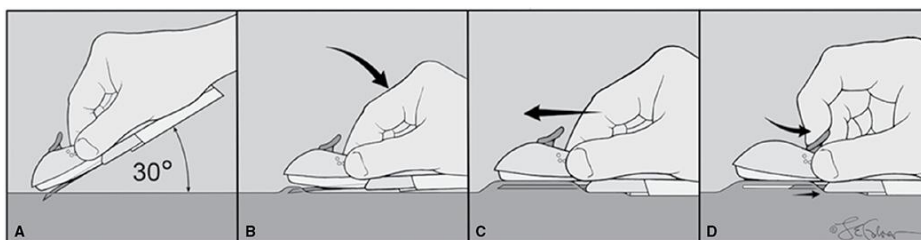


Figure 38-2. Nexplanon insertion: (A) enter at 30°, (B) drop to horizontal, (C) tent the skin while advancing subdermally, (D) pull the lever to retract the needle and deposit the implant.

Implant removal

- Antiseptic prep → press the **proximal end** so the distal end bulges → anesthetize → **2-mm incision toward the elbow** along the long axis → push the proximal end toward the incision → grasp the distal end with a hemostat; dissect superficial adhesions.

Injectable progestins

Agent	Dose / interval	Notes
DMPA IM (Depo-Provera)	150 mg every 3 months	Deltoid or gluteus; do not massage (slow release)
Depo-subQ provera 104	Every 3 months SC (anterior thigh or abdomen)	Can be self-administered
Norethisterone enanthate	200 mg every 2 months IM	Used worldwide (not US)

- **First injection within 5 days of menses onset** → contraceptive levels by **24 h**, no back-up needed.
- **Quick start** (any cycle day): negative pregnancy test first, **back-up for 7 days**, repeat pregnancy test at **3–6 wk**.
- Pregnancy on DMPA → **no ↑ anomalies**.

DMPA-specific adverse effects

- **Delayed return of fertility** (prolonged anovulation): **¼ lack regular menses for up to 1 year** → poor choice for brief use before planned conception.
- No ↑ CV events/stroke in healthy women, but **↑ stroke with severe hypertension**. US MEC concern for **hypoestrogenism and ↓ HDL** in vascular disease/multiple CV risks.
- **Weight gain 1–3 kg in year 1**.
- **Bone density loss** in long-term users, **slowly reversible** after stopping.
 - **FDA black box warning (2004)** — most relevant to **adolescents** and **perimenopausal** women — but **DMPA should NOT be restricted** in these groups (ACOG).

Progestin-only pills ("mini-pills")

Pill	Dose / regimen	Ovulation inhibited?	Missed-pill window
Norethindrone	0.35 mg daily, continuous	Not reliably (mucus + endometrium)	3 h → back-up for 48 h
Drospirenone (Slynd)	4 mg, 24 days of 28 (4 hormone-free days ↓ irregular bleeding)	Yes	24 h (like COCs)
Desogestrel	0.075 mg daily, continuous (not in US)	Yes	12 h

- Norethindrone pills must be taken at the **same time daily** (mucus effect is short-lived).
- **Drospirenone** (spironolactone analog): antiandrogenic, antialdosterone (↓ water retention), antimineralocorticoid → theoretical **hyperkalemia**.
 - **Contraindicated in renal or adrenal insufficiency** (plus usual progestin contraindications).
 - **Check K⁺ in the first month** if also on NSAIDs, ACE inhibitors, ARBs, heparin, aldosterone antagonists or K⁺-sparing diuretics.

4. Combination Hormonal Contraceptives (CHCs)

Mechanism

- **Main effect: suppression of hypothalamic gonadotropin-releasing factors** → **no ovulation** (both components).
- Progestin: thick mucus, unfavorable endometrium.
- **Estrogen: stabilizes the endometrium** → **prevents breakthrough bleeding ("cycle control")**.
- Extremely effective yet highly reversible.

Combination oral contraceptive pills — composition

- **Ethinyl estradiol (EE)** is the usual estrogen; less often mestranol, estetrol, estradiol valerate.
- Estrogen side effects: **breast tenderness, weight gain, nausea, headache**.

Table 38-5. Estrogen and Progestins Used in FDA-Approved Combination Oral Contraceptive Pills

Estrogen Type (Pill Dose)	Paired Progestin ^a
Ethinyl estradiol 10 µg	Norethindrone acetate
Ethinyl estradiol 20 µg	Desogestrel, Drospirenone, Levonorgestrel, Norethindrone acetate
Ethinyl estradiol 25 µg	Desogestrel, Norgestimate
Ethinyl estradiol 30–35 µg	Desogestrel, Drospirenone, Ethynodiol diacetate, Levonorgestrel, Norethindrone, Norethindrone acetate, Norgestimate, Norgestrel
Ethinyl estradiol 50 µg	Ethynodiol diacetate, Norgestrel
Estradiol valerate 2 mg	Dienogest
Estetrol 14.2 mg	Drospirenone
Mestranol 50 µg	Norethindrone

^aProgestins are listed alphabetically. FDA = U.S. Food and Drug Administration.

- Progestins are related to **progesterone, testosterone or spironolactone** → variable binding to progesterone, androgen, glucocorticoid and mineralocorticoid receptors (explains side effects); clinical superiority of one over another is unclear.
- **EE 10–50 µg/day; most ≤35 µg**. The floor is set by pregnancy prevention + cycle control.
- **"Fe"** pills: iron replaces placebo. **Beyaz, Sayfral: levomefolate** (folate) in active and placebo pills.
- **Monophasic** (constant dose) vs **bi-/tri-/quadriphasic**: phasic pills aim to ↓ total progestin, but **no proven clinical advantage**; cycle control similar.

Administration

- Active pills **21–81 days** + placebo **4–7 days** (pill-free interval → withdrawal bleed).
- **24/4 regimens** counter follicular growth with low-dose pills; perform like **21/7**.
- **Extended cycle**: 12 wk active + 1 wk off (**13-wk cycle**). **Amethyst**: continuous **365 days**. Same efficacy/safety; good for significant menstrual symptoms.

Start method	Back-up needed?
First day of menses	No
Sunday start (first Sunday after menses begins)	Yes, 1 wk (none if menses begin on Sunday)
Quick start (any day, usually day prescribed)	Yes, 1 wk

- **COCs are not teratogenic** if the woman is unknowingly pregnant. A missed menses after starting → pregnancy test. Same start rules for rings and patches.

Missed pills	Action
1 dose	Take missed pill now + today's pill on time; continue
≥2 doses	Take most recent missed pill now + today's on time; back-up for 7 days ; continue
No withdrawal bleed in pill-free interval	Continue pills, exclude pregnancy

- Early spotting is common, resolves in **1–3 cycles**, and is **not** method failure.
 - Persistent bleeding **early in the pack** → ↑ **estrogen**; **late in the pack** → ↑ **progestin**.

Drug interactions

- **↓ COC efficacy (choose another method):**
 - **Rifampin/rifabutin** (other antibiotics are **not** implicated)
 - **CYP450-inducing anticonvulsants:** phenytoin, phenobarbital, primidone, carbamazepine, oxcarbazepine, topiramate, rufinamide, felbamate
 - Some antiretrovirals (mainly **protease inhibitors**)
- **COCs ↓ lamotrigine** efficacy.
- **COCs remain effective in obesity;** the **Ortho Evra patch** may be less effective.

Metabolic changes, risks and benefits

Table 38-6. Contraindications to the Use of Combination Oral Contraceptive Pills

Contraindication
Pregnancy
Uncontrolled hypertension
Smokers older than 35 years
Diabetes with vascular involvement
Cerebrovascular or coronary artery disease
Migraines with associated focal neurologic deficits
Thrombophlebitis or thromboembolic disorders
History of deep-vein thrombophlebitis or thrombotic disorders
Thrombogenic heart arrhythmias
Thrombogenic cardiac valvulopathies
Undiagnosed abnormal genital bleeding
Known or suspected breast carcinoma
Cholestatic jaundice of pregnancy or jaundice with pill use
Hepatic adenomas and carcinomas
Cirrhosis or active liver disease with abnormal liver function
Known or suspected estrogen-dependent neoplasia

Hypertension

- COCs ↑ hepatic **angiotensinogen** → possible pill-induced hypertension → **recheck BP 8–12 wk** after starting. Low-dose pills rarely cause significant hypertension.
- **Well-controlled hypertension → US MEC 3** (↑ stroke, MI, PAD). **Severe hypertension / end-organ damage → avoid.**

Venous thromboembolism

- Estrogen ↑ fibrinogen and clotting factors; risk is **estrogen-dose related**.
- Baseline VTE **4–5 per 100,000 woman-years**; **COCs ↑ 3–5-fold**.
- **Obesity → US MEC 2** (compounds risk). **Smokers >35 → not recommended**. **Thrombophilias** carry the highest risk.
- COC use in the month before **major surgery ~doubles postoperative VTE** → weigh stopping **4–6 wk before surgery** (time to reverse the thrombogenic effect) against pregnancy risk.
- **Early puerperium → avoid** (Table 38-4).
- **Progestin type: drospirenone, desogestrel, gestodene** → slightly higher VTE risk.

Arterial disease

- **Prior MI → do not use.** Multiple CV risks (smoking, hypertension, older age, dyslipidemia, diabetes) → MI risk outweighs benefit. No CV risks → **no ↑ MI** with low-dose pills.
- **Stroke:** rare in nonsmokers <35; small ↑ ischemic stroke. **Much higher with hypertension, smoking, or migraine with aura/focal signs.**
 - Migraine **without aura:** may start if young, healthy, normotensive nonsmoker. **Prior stroke → avoid.**

Carbohydrate and lipids

- **No ↑ diabetes.** Diabetics may use COCs if **nonsmokers, disease <20 yr, no vascular disease, nephropathy, retinopathy or neuropathy.**
- ↑ TG, total and HDL cholesterol; ↓ LDL. **No weight gain.**

Neoplasia	COC effect
Ovarian and endometrial cancer	Protective
Colorectal cancer	↓ in ever users
Cervical dysplasia/cancer	↑ in current users ; returns to baseline ≥10 yr after stopping
Breast cancer	Unclear — none or small ↑ in current users, fading after cessation
Liver tumors	No ↑ risk. OK with focal nodular hyperplasia ; avoid with hepatic adenoma and HCC

Liver disease

- Cholestasis/cholestatic jaundice: uncommon, resolves on stopping.
- **Active hepatitis:** don't **start** COCs (may **continue** if a flare happens while on them); progestin-only unrestricted; fine after recovery.
- **Mild compensated cirrhosis:** no restriction. **Severe decompensated cirrhosis: avoid all hormonal methods.**
- **Chloasma:** more likely if it occurred in pregnancy; less with low-dose pills.

Noncontraceptive benefits

- ↓ **dysmenorrhea and heavy menstrual bleeding.**
- ↑ **SHBG** → binds testosterone → ↓ **acne and hirsutism.**
- **Drospirenone COCs** ↓ **PMDD** symptoms.
- ⚠ **Low-dose COCs do not treat or prevent functional ovarian cysts.**

Intravaginal rings

Feature	NuvaRing	Annovera
Hormones	EE + etonogestrel	EE + segesterone acetate
Size	54 mm diameter, 4 mm cross-section (clear)	56 mm, 8.4 mm (white)
Use	3 wk in, 1 wk out; new ring each cycle	3 wk in, 1 wk out; wash and reuse — 13 cycles per ring
Out for intercourse	Replace within 3 h	Replace within 2 h

- Insert within **5 days of menses onset**; no special orientation.
- High satisfaction; **more vaginitis, ring events and leukorrhea** than COCs.
- Compatible with tampons and vaginal medications — **except Annovera + miconazole suppositories** (↑ hormone levels) → use cream or oral antifungals.

Transdermal patches

- **Ortho Evra / Xulane:** **EE + norelgestromin.** **Twirla:** **EE + levonorgestrel.**
- Apply to buttocks, upper outer arm, lower abdomen or upper torso — **not the breasts.**
- **New patch weekly × 3, then 1 patch-free week.** Replace if it needs tape. Saunas/whirlpools OK; **Twirla:** **swimming ≤30 min.**

- Efficacy, side effects and metabolic effects similar to COCs, but:
 - **Ortho Evra: possible ↑ VTE** (label RR **1.2–2.2**) and **↑ failure at ≥90 kg**.
 - **Twirla**: VTE comparable to COCs; efficacy similar in obese and normal-weight women.

5. Barrier Methods

Male condom

- Considerable (not absolute) **STD/HIV protection**.
- Efficacy ↑ with **reservoir tip + spermicide**. **Use water-based lubricants** (oil degrades latex).
- **Lambskin**: effective contraception but **no infection protection**. **Polyurethane/synthetic elastomer**: STD-protective but **more breakage and slippage** than latex.

Female condom (FC2)

- **Nitrile sheath** + outer ring + flexible **polyurethane inner ring**. Open ring outside the vagina; **closed inner ring under the symphysis** like a diaphragm. Silicone-lubricated.
- ⚠ **Never use with a male condom** (friction → slipping, tearing).
- Twist the outer ring after use to keep semen in. Some STD protection.

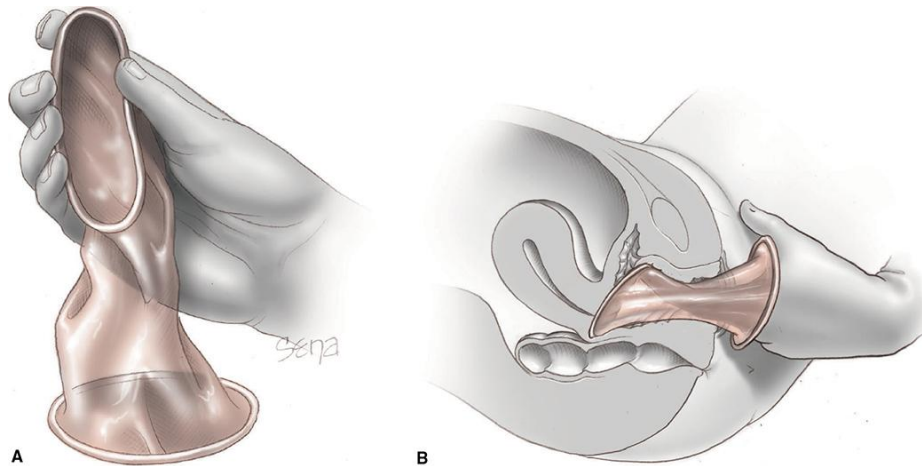


Figure 38-3. FC2 female condom: (A) squeeze the inner ring to insert; (B) push the inner ring up behind the symphysis with a finger.

Diaphragm + spermicide

- Latex dome on a spring rim; **water-based spermicide** in the cup and along the rim. Cup faces the cervix.
- Correct position: one rim in the **posterior fornix**, the other **behind the symphysis, just below the urethra**. Too small → dislodges; too large → uncomfortable. **Prolapse** → expulsion. **Fitted and prescribed**.
- Can be inserted hours ahead; **>6 h before sex → add spermicide**; add spermicide **before each act**.
- **Leave in ≥6 h after intercourse**; remove at 6 h or next morning (rare **TSS**).
- Slightly ↑ **UTIs** (rim irritating the urethra).

Cervical cap (FemCap)

- Silicone "sailor cap"; sizes **22, 26, 30 mm**; spermicide once on both sides of the dome.
- Leave **≥6 h after coitus**, up to **48 h**. **Pregnancy rates higher than with the diaphragm**.

6. Fertility Awareness–Based Methods (FABMs)

- Identify fertile days and abstain. **Typical-use pregnancy ~15%**. Smartphone apps may help.

Method	Rule
Standard Days	Avoid unprotected sex on cycle days 8–19 ; needs regular cycles of 26–32 days (CycleBeads)
TwoDay	Sex considered safe if no cervical mucus today or yesterday
Symptothermal	Mucus marks the start , a sustained 0.4°F BBT rise marks the end of the fertile window, plus calculations. Abstain from day 1 of menses to the 3rd day after the temperature rise . More complex; not appreciably more effective.

The chapter states that the BBT rise "usually precedes ovulation"; physiologically the progesterone-driven rise follows ovulation, which is why it marks the end of the fertile period.

7. Spermicides

- Mostly **nonoxynol-9** (OTC creams, jellies, suppositories, films, foams): chemical spermicide + physical barrier. **Less effective; max effect ≤ 1 h; no STD protection.** Not teratogenic.
- Phexxi (2020)**: vaginal gel of **lactic acid, citric acid, potassium bitartrate** → keeps vaginal pH acidic despite semen buffering (possible microbicide).
- Place high in the vagina **up to 1 h before coitus**; reapply for each act.

Contraceptive sponge (Today)

- OTC, one size; **nonoxynol-9** polyurethane disc **2.5 cm thick, 5.5 cm wide**; dimple faces the cervix, loop for removal.
- Insert **up to 24 h before**; works regardless of coital frequency; **leave ≥ 6 h after**.
- Works mainly by spermicide (also covers cervix, absorbs semen). **Less effective than diaphragm or condom.**
- Stopped mainly for pregnancy, discomfort, vaginitis. TSS rare (sponge may even limit toxin), but **avoid during menses and the puerperium**.

8. Emergency Contraception (EC)

Table 38-7. Emergency Contraception Methods and Some Brand-Name Examples

Method	Formulation	Pills per Dose	Number of Doses ^a
Progestin-only Pill			
Plan B One-Step	1.5 mg LNG	1	1
PRM Pill			
Ella	30 mg ulipristal acetate	1	1
COC Pills^b			
Ogestrel	50 µg EE + 0.5 mg norgestrel ^c	2	2
Cryselle, Low-Ogestrel	30 µg EE + 0.3 mg norgestrel ^c	4	2
Enpresse (orange), Trivora (pink)	30 µg EE + 0.125 mg LNG	4	2
Levora, Seasonale	30 µg EE + 0.15 mg LNG	4	2
Aviane, LoSeasonique (orange)	20 µg EE + 0.1 mg LNG	5	2
Copper-containing IUD			
Paragard T380A			

^aDoses taken 12 hours apart. ^bCOCs method, also known as the Yuzpe method, aims to provide a minimum of 100 µg of EE and 0.5 mg of LNG or 1.0 mg of norgestrel in each of the two doses. ^cNorgestrel contains two isomers, and only one of these isomers is bioactive, namely levonorgestrel. Thus, the amount of norgestrel needed for efficacy is

twice that of the levonorgestrel-based regimens. COC = combination oral contraceptive; EE = ethinyl estradiol; IUD = intrauterine device; LNG = levonorgestrel; PRM = progesterone-receptor modulator. The printed footnote reads "aims to provides"; corrected to "provide".

- **Single-dose LNG is OTC** for all reproductive-aged people (888-NOT-2-LATE; not-2-late.com).
- **LNG-IUS was noninferior to Cu-IUD for EC** in early data (not yet adopted).

Hormonal EC

- **No US MEC contraindication** except allergy. **No exam or pregnancy test needed.**
- Start **ASAP, ideally ≤ 72 h**; may be given up to **120 h**.
- **Main mechanism: inhibition or delay of ovulation.**
- **Failure: ulipristal lowest (1–2%); Yuzpe highest (2–3.5%).**
- If it fails, **no $\uparrow$ major malformations** or pregnancy complications.
- **Nausea/vomiting:** antiemetic **≥ 1 h before** each dose; **vomiting within 2 h $\rightarrow$ repeat the dose.**
- Afterward use a barrier method until long-term contraception starts; EC may be repeated within a cycle. Quick-start other methods is fine.
- **⚠ After ulipristal, wait 5 days before starting a progestin-containing method** (competes at the progesterone receptor).

Copper IUD for EC

- **The most effective EC**, and gives **10 years** of contraception.
- Inserted **up to 5 days** after unprotected sex $\rightarrow$ pregnancy **$\sim 0.1\%$** .

Quick-Recall: Method Comparison

Method	Key mechanism	Duration / regimen	Typical failure	Signature caveats
Cu-IUD (ParaGard)	Sterile inflammation; no ovulation block	10 yr	0.8%	Heavy bleeding; Wilson disease; best EC (≤ 5 d)
LNG-IUS 52 mg (Mirena/Liletta)	Atrophy + mucus	5 / 6 yr	0.1%	Amenorrhea 35% at 3 yr; breast cancer contraindicates
Implant (Nexplanon)	Ovulation suppression	3 yr	0.1%	Insert 8–10 cm above medial epicondyle; nerve injury if deep
DMPA	Ovulation suppression	150 mg IM q3mo	4%	Delayed fertility, 1–3 kg gain, bone loss (black box but not restricted)
POP — norethindrone	Mucus + endometrium	0.35 mg daily	7%	3-h window
POP — drospirenone (Slynd)	Ovulation block	4 mg, 24/4	7%	24-h window; K ⁺ ; avoid renal/adrenal insufficiency
COC	Ovulation block (GnRH suppression)	21/7, 24/4, extended, continuous	7%	VTE $\times 3$ –5; no estrogen < 21 d postpartum; smokers > 35 ; migraine with aura
Ring	As COC	3 wk in / 1 out	7%	NuvaRing 3 h / Annovera 2 h out; Annovera 13 cycles
Patch	As COC	Weekly $\times 3$	7%	Ortho Evra: $\uparrow$ VTE, fails ≥ 90 kg
Male condom	Barrier	Each act	13%	STD protection; water-based lube
Diaphragm	Barrier + spermicide	Leave ≥ 6 h	24%	UTI, TSS; fitted
Spermicide	Nonoxynol-9 / acid gel	≤ 1 h before	28%	No STD protection

Key Numbers to Memorize

- Exclusive breastfeeding: ovulation unlikely for **10 wk**; ~½ of US pregnancies unintended
- Typical failure: LARC **0.1–0.8%**, DMPA **4%**, pill/patch/ring/POP **7%**, male condom **13%**, female condom **21%**, diaphragm **24%**, spermicide **28%**, FABM **15%**
- IUD lifespans: Mirena **5**, Liletta **6**, Kyleena **5**, Skyla **3**, ParaGard **10 yr**
- IUD: expulsion **~10%** at 3 yr (most in month 1); strings check **4–6 wk**; perforation **1/1000**; infection risk highest first **3 wk**; strings trimmed to **3–4 cm**; PID not improving in **48–72 h** → remove
- Pregnancy with Cu-IUD: abortion **54% vs 25%**, preterm **17% vs 4%** (left vs removed)
- Immediate IUD = **<10 min** after placenta; early = 10 min–4 wk (highest expulsion)
- CHCs postpartum: **<21 d = category 4**
- Nexplanon **≥30 µg/day × 3 yr**, 4-cm rod, insert **8–10 cm** above medial epicondyle, bevel at **30°**; complications **1%** insertion / **5%** removal
- DMPA **150 mg IM q3mo**; ¼ no menses for **1 yr** after; weight gain **1–3 kg**; quick start → back-up **7 d**, pregnancy test **3–6 wk**
- POP windows: norethindrone **3 h** (back-up 48 h), desogestrel **12 h**, drospirenone **24 h**
- COC: EE **10–50 µg** (most ≤35); VTE baseline **4–5/100,000** → **×3–5**; stop **4–6 wk** before major surgery; BP check at **8–12 wk**; diabetes OK if **<20 yr** duration without vascular disease; cervical cancer risk normalizes **≥10 yr** after stopping
- Standard Days: avoid days **8–19** (cycles 26–32 d); BBT rise **0.4°F**
- EC: **≤72 h** ideal, up to **120 h**; ulipristal failure **1–2%**, Yuzpe **2–3.5%**; repeat if vomiting within **2 h**; wait **5 d** after ulipristal to start progestin; Cu-IUD **≤5 d** → **0.1%** pregnancy; Yuzpe **≥100 µg EE + 0.5 mg LNG** per dose, 2 doses **12 h** apart